Pipe flow on one page

1 Basics

$Q = VA$

$Re = \frac{VD}{\nu}$: laminar below 2,300; turbulent above about 4,000

not round: $D_h = \frac{4A}{P}$

velocity head $\frac{V^2}{2g}$

$\nu = \frac{2.414 \times 10^{-5} \cdot 10^{247.8/(T+133.15)}}{1000}$ (T in °C)

new pipe ε (mm): cast iron 0.26, steel 0.046, plastic 0.0015

2 Energy

$z_1 + \frac{p_1}{\gamma} + \frac{V_1^2}{2g} + h_p = z_2 + \frac{p_2}{\gamma} + \frac{V_2^2}{2g} + h_t + h_L$

$\mathrm{EGL} = z + \frac{p}{\gamma} + \frac{V^2}{2g}, \quad \mathrm{HGL} = z + \frac{p}{\gamma}$

pressure head = HGL − pipe level

power $P = \frac{\gamma Q H}{\eta}$ (grid: pump η × motor η)

3 Friction

$h_f = f \frac{L}{D} \frac{V^2}{2g}$

laminar: $f = \frac{64}{Re}$

Colebrook: $\frac{1}{\sqrt{f}} = -2 \log\left(\frac{\varepsilon}{3.7D} + \frac{2.51}{Re\sqrt{f}}\right)$

Swamee-Jain: $f = \frac{0.25}{\left[\log\left(\frac{\varepsilon}{3.7D} + \frac{5.74}{Re^{0.9}}\right)\right]^2}$

Hazen-Williams, SI, water only: $h_f = \frac{10.67\, L\, Q^{1.852}}{C^{1.852}\, D^{4.87}}$

4 Minor losses

$h_m = K \frac{V^2}{2g}$

square entrance 0.5; exit 1.0; elbows 0.4 to 0.9
open valves: gate 0.2; swing check 2.5; globe 10

sudden expansion $h_L = \frac{(V_1 - V_2)^2}{2g}$

contraction $K = 0.5\left(1 - \frac{A_2}{A_1}\right)$, small-pipe V

equivalent length $L_e = \frac{\sum K \cdot D}{f}$

5 Pipe systems

between two surfaces: $z_1 - z_2 = \left(f \frac{L}{D} + \sum K\right) \frac{V^2}{2g}$

series: the same Q; losses add
parallel: the same loss; flows add

free jet: add one $\frac{V^2}{2g}$, no exit loss

6 Meters and pumps

$Q = C_d\, A_2 \sqrt{\frac{2\, \Delta p/\rho}{1 - \beta^4}}, \quad \beta = \frac{d}{D}$

Cd about 0.98 venturi, 0.96 nozzle, 0.61 orifice plate

Pitot: $V = \sqrt{2g\, \Delta h}$

pump runs where $h_{\mathrm{pump}}(Q) = H_s + \left(f \frac{L}{D} + \sum K\right) \frac{V^2}{2g}$

$\mathrm{NPSH}_a = \frac{p_{\mathrm{atm}} - p_{\mathrm{vap}}}{\gamma} - z - h_s$, above NPSH required

What is unknown?

loss hf
Darcy-Weisbach directly

flow Q
repeat from the fully rough f, or Swamee-Jain

diameter D
try sizes or Solve; round up; check V

several pipes
series and parallel rules

a pump
the crossing, then η and NPSH

SI units. K and roughness: US EPA, EPANET 2.2 User Manual, Tables 3.2 and 3.3 (public domain). Pages 2 to 9: each part in large print. Free to share with credit.

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Choosing the method

What is unknown?

loss hf	→ Darcy-Weisbach directly
flow Q	→ repeat from the fully rough f, or Swamee-Jain
diameter D	→ try sizes or Solve; round up; check V
several pipes	→ series and parallel rules
a pump	→ the crossing, then η and NPSH

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Five habits

- ✓ SI units everywhere
- ✓ Check Re before choosing f
- ✓ Each K with its own velocity
- ✓ Round a diameter up, never down
- ✓ Follow the HGL along the whole pipe

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